

The economics of carbon capture, use and storage

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July 2025

Abstract

Text of the Abstract.

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Keywords: Text

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1 Introduction

What is the motivation for developing this technology?

Carbon, capture, utilisation, and storage (CCUS) technologies are important tools to reduce CO₂ emissions arising from big industrial sources and the power sector. The CCUS value-chain covers from the capture of carbon dioxide â either at the source of emissions or from the atmosphere â to its ultimate storage, burial, or usage, preceded by its adequate transportation. Often, carbon capture technologies target emission at its source. Most commonly these are industrial processes, power stations, refineries, and cement production facilities. However, direct air capture technologies can also extract carbon directly or indirectly from the atmosphere, possibly long after the original emission (IEA, 2023a).

The role of CCS in decarbonisation is four-fold. First, they reduce emissions from existing power and industrial infrastructure, as they can be retrofitted to existing industrial facilities (namely heavy industry) or fossil fuel-based power plants. Second, CCUS technologies support the production of low-carbon hydrogen, which has growing demand in transportation, industry, and construction. Third, CCUS technologies are an avenue to tackle the most challenging emissions in heavy industries, such as the ones arising from cement production and iron, steel, and chemicals manufacturing. Finally, direct air capture technologies (DAC), also known as carbon removal technologies remove carbon from the atmosphere that results from emissions. DAC technology is the most direct and measurable way to remove carbon from the atmosphere. We will discuss more below, but there is a strong reliance on this technology in the policy modelling of decarbonisation and net zero (both at the country level and for individual firms).

In order to meet deep decarbonisation targets by 2050, CCS will have to play a role in abatement of emissions. The International Energy Association (IEA) has CCS playing a vital role in their Sustainable Development Scenario (SDS) as well as the more ambitious Net Zero Scenario (NZE) model: "Meeting net zero goals will require widespread deployment of carbon capture, utilisation and storage (CCUS). CCUS is the only group of technologies that can contribute to deep emission reductions in key sectors, including heavy industry, while also supporting the removal of CO₂ from the atmosphere.â The IEA modelling for emissions in the NZE scenario rely explicitly on the development and deployment of CCS for the abatement of energy sector emissions. The NZE scenario calls for 1,286 Mt CO₂ capacity of CCS, while today there is only 44 Mt CO₂ capacity of CCS projects operating (IEA, 2023a). Without CCS, it is very unlikely deep emissions reductions in the energy sector can be met.

The Intergovernmental Panel on Climate Change (IPCC) regularly produces descriptive climate and policy scenarios using an integrated assessment model (IAM) approach, where achieving mitigating targets relies heavily on the development of CCS. In the first set of IAM models produced by the IPCC, the Representative Concentrated Pathways (RCP), CCS deployment is critical in allowing for overshoot scenarios - where peak emissions exceed targets, but decrease over time to meet the carbon budget target by 2100 ([Van Vuuren et al., 2011](#)). After the release of the Shared Socioeconomic Pathways (SSPs) in the 5th Annual Report, the IPCC model reported that to achieve the target of 450 ppm of CO₂, CCS technologies in tandem with biomass was essential ([Tavoni and Socolow, 2013](#)). For over a decade, CCS has featured as a critical technology to achieving global decarbonisation targets, yet, the technology remains uncertain today.

Within the last five years, government support has ramped up to meet climate ambitions. In the western world, a wave of green industrial policies has mobilised billions of dollars into research and development, deployment and scaling of low-emissions technologies. Much of this pot has been funnelled to CCS, including in the US and the UK. In the UK, a £1 billion fund into CCUS transportation and storage was enacted following the government's 10-Point Green Industrial Plan announced in 2020 ([Government, 2022](#)). The fund will primarily support transportation and storage infrastructure and will focus on deployment of CCUS for industrial and power sector use in the first phase. Additionally, £20 billion over twenty years for CCUS was pledged in the Spring 2023 budget by the UK government ([Burke, 2023](#)). This funding will focus on reducing capital cost to create industrial clusters with CCUS ([Velev, 2023](#)). The UK pledged £100 mn for DAC technologies, while in the US, the Inflation Reduction Act (IRA) has created a \$3.5 bn fund to develop DAC technologies, and an additional \$180 credit per ton of carbon ([Lo, 2023](#)). These incentives are designed to lower the technological cost and accelerate deployment for climate targets.

Developed economies have an incentive to innovate in carbon emissions reductions technologies because of their large share of current and historical emissions. Innovating in CCS and CCUS technology is important because it is the only mitigation intervention which reduces net emissions under any quantity of emissions output. It is the only engineering solution for carbon dioxide removal which can store carbon for up to hundreds of years after removal ([Gibbs et al., 2023](#)). While there's no keyhole to peer through into the future, international climate policy could shift to ensure nations with high historical emissions account for their use of the carbon budget. Having CCUS technology at a scale would become a critical way to compensate for historical emissions. At this point, CCUS is not scalable

for deployment to meet IPCC scenario modelling, let alone make up for decades of previous emissions.

The private sector also has an incentive to create cost competitive CCS technologies. Growth in sector is driven by demand by corporates and industry to reduce firm-level emissions, and in some cases, create a new revenue stream. Many firms are voluntarily disclosing and creating ambitions to lower their own emissions. Even firms without direct emissions, such as banks, want to reduce the emissions intensity of their investments and are doing so through collaborative efforts with downstream industries. For example, the *Glasgow Financial Alliance for Net Zero (GFANZ)*, an alliance of financial sector firms, acknowledges the role banks will need to play in mobilizing funding and encouraging real economy sectors to undertake emissions reductions ([GFANZ, 2022](#)). To meet voluntary targets, firms across sectors are working to reduce their overall emissions, and negative emissions technologies are a high reward investment if achievable.

Firms also participate in the voluntary trade of carbon offsets, where emissions reductions are purchased and sold to meet firm-level commitments. This market is predicted to grow substantially in the next ten years ([Porsborg-Smith et al., 2023](#)). Carbon credits are often considered of higher value - subsequently fetching a higher market price - when they create new emissions reductions. CCS and CCUS projects are particularly well suited to creating additional emissions reductions, beyond what policy and market scenarios already generate. This makes CCS a valuable investment for the future as the cost per tonne of carbon sequestered is considerably higher than other types of credits, like energy efficiency upgrading.

Private support for CCS is currently non-optimal, due to both typical market failures at early stage innovation and the disparate externality of global carbon dioxide emissions. The issue is, given the state of innovation in the industry, risk has kept private investment below the socially optimal amount. Despite these challenges, there is a small, but secure source of demand for the technology. Stripe Climate has created a platform to vet future carbon offsets, called Frontier, for thousands of companies looking to buy high quality carbon removal credits from CCS and CCUS projects. Companies such as Meta, McKinsey Sustainability and Alphabet have used Advanced Market Commitments (AMC) to pledge future purchases of these credits([FrontierClimate, 2023](#)). This funding arrangement has allowed innovators to secure private funding for improving the feasibility of different technologies at scale. This innovation is supported by today's growing demand for this technology.

2 Technical components

Under the umbrella of carbon capture, use and storage there are different mechanical and technical processes. Below we review the different components of carbon reuse and removal technologies to better understand their application. Each component, either alone or in tandem with the another process, have a unique end-user. Understanding the components of these end-use technologies and their application disentangles the sector into more clear technological processes. Each process has a unique use case, and improving the cost efficiency of one or joint components will have varying impacts for the downstream end use sector.

We will look at each component along the process - including capture, transportation, storage and utilisation and provide a brief technical overview.

2.1 Carbon capture

These technologies mainly aim to capture CO₂ from very large sources (power plants, gas processing facilities, industrial processes). The most recent developments in carbon capture technologies also target extraction from the atmosphere (direct air capture).

Table 1: Overview of Capture Technologies, adapted from IEA (2021)

Technology	Overview	Implementation
<i>Chemical Absorption</i>	A common process based on the reaction between CO ₂ and a chemical solvent (e.g., amines like ethanolamine). This is the most advanced CO ₂ separation technique.	Widely used for decades, applied in power generation, fuel transformation, and industrial production.
<i>Physical Separation</i>	Based on either adsorption, absorption, cryogenic separation, or dehydration. Adsorption uses activated carbon, alumina, or zeolites.	Mostly used in natural gas processing, ethanol, methanol, and hydrogen production, with nine projects in operation.

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Technology	Overview	Implementation
<i>Oxy-Fuel Separation</i>	Involves burning fuel in nearly pure oxygen, with subsequent capture of the CO ₂ emitted. This results in a high-purity CO ₂ stream.	At large pre-demonstration stage. Some projects completed in coal-based power and cement production.
<i>Membrane Separation</i>	Uses polymeric/inorganic membranes with high CO ₂ selectivity, allowing CO ₂ to pass but blocking other gases.	Varies by fuel type. Petrobas operates the only large-scale capture plant using this method.
<i>Calcium Looping</i>	Uses CaO as a sorbent to capture CO ₂ , forming calcium carbonate (CaCO ₃), which is then regenerated in a second reactor.	Pilot/pre-commercial stage, tested in fluidized bed combustors and cement manufacturing.
<i>Chemical Looping</i>	Uses metal oxides (e.g., iron/manganese) to bind O ₂ for controlled combustion, producing concentrated CO ₂ .	Tested in 35 pilot projects in coal, gas, oil, and biomass.
<i>Direct Separation</i>	Captures CO ₂ from limestone heating using special calciners, avoiding mixing with combustion gases.	Being tested in Low Emissions Intensity Lime and Cement (LEILAC) in Belgium.
<i>Supercritical CO₂ Power Cycles</i>	Uses CO ₂ instead of steam as the working fluid, improving efficiency.	Two prototype projects: NET Power's Allam cycle and Trigen CES.

The technologies seeking to remove CO₂ released in the burning of fossil fuels at power plants can be classified as post-combustion, pre-combustion or oxyfuel combustion technologies. Post-combustion techniques remove CO₂ after combustion from the exhaust gas (scrubbing). They are already well developed and frequently used in industrial applications. Pre-combustion techniques capture CO₂ before the fossil fuel is burnt by pre partially burning fuel and forming a synthetic gas, with hydrogen as a by product. Carbon monoxide is then extracted from this exhaust gas stream. Pre-combustion carbon capture techniques are used in power production, chemical gasfuel production and fertiliser production.

Finally, oxyfuel combustion techniques rely on burning the fossil fuel in oxygen, rather than in air. Combustion results in a flue gas formed mainly by CO₂ and water vapour. Once

the water vapour condenses, the almost pure CO_2 is stored. Oxyfuel combustion allows capturing almost all the CO_2 . However, it is an energy intensive process.

Direct air capture techniques are much more novel and still being researched. DAC processes extract the CO_2 from the atmosphere, and then stores the CO_2 in geological formations (IEA, 2023c). There are two distinct types of capturing carbon from the air, including bioenergy production with CCS (BECCS) and DAC, as discussed above. With DAC technology, there are different processes to sequester carbon directly, discussed in more detail in the next section.

BECCS is unique in that biomass is harvested and used as an input for energy production in a plant which stores the carbon emissions from that generated energy (Herzog, 2023). This process then takes the naturally captured CO_2 from photosynthesis in the burnt biomass and stores it in the ground, resulting in capturing and storage of emissions. DAC, distinctly, uses a chemical process to remove concentrations of CO_2 from the air. These techniques are slightly more energy intensive than capturing carbon dioxide at emission sources and are currently more costly. Currently, only eighteen DAC facilities exist worldwide, but there is growing interest as capture rates improve. The costs remain high, and innovation in this capture technology is still early stage (IEA, 2023c).

2.2 Carbon transportation

Carbon transportation is the process of moving sequestered or captured carbon to another location for use or storage. The transport of carbon is not a completely new process, given its application for use in oil and gas for decades. Despite the demonstration of this technology, the cost of CO_2 transport is extremely variable. Both the quantity and distance contribute to the cost of transportation infrastructure (McCoy and Rubin, 2008). Carbon can be transported in three states: gas, liquid and solid (Doctor et al., 2018).

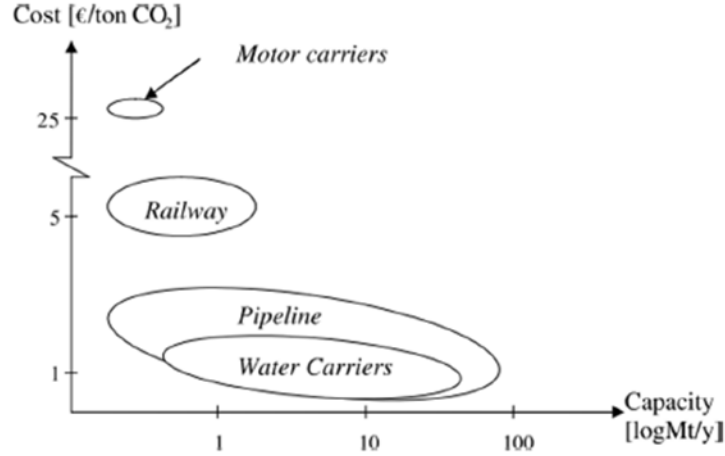


Fig. 2. Cost and capacity for transportation alternatives at 250 km [26].

Figure 1: Caption

Carbon has several safe modes of transportation, derived from oil industry best practices. Beginning with the most cost effective, carbon can be transported via pipeline, small scale shipping, barge, train and truck (IEA, 2023d). Current modes of transportation were developed and scaled primarily for use in the oil and gas sector. In the future, transportation methods will need to be developed with storage as the end use.

- Pipelines are the most common and cost effective method for transportation. Pipeline infrastructure already exists across the US, Canada and Europe, primarily alongside oil and gas extraction. As of 2022, 70% of current pipelines transporting 35 Mt of CO₂ per year for carbon are dedicated to enhanced oil recovery (EOR) (IEA, 2023d). There exist about 6,500 km of pipelines dedicated to the transportation of carbon in the US (Institute, 2018), with about 9,000 km existing globally IEA (2023d). Pipeline infrastructure requires economy of scale for production, so pipelines are ideal for the development of large scale and network-based transportation of CO₂ (Svensson et al., 2004).
- Pipeline transportation will likely continue to be the most common type of carbon transport. Both pipelines and ships can transport CO₂ as gas or liquid. Currently, gaseous transport is most common, although given the volume of gas, large facilities are therefore required to compress and transport CO₂ (Doctor et al., 2018).
- Pipeline transportation operate on both land and sea. Capital expenditures on the transmission pipeline itself are the driver of the cost - making new pipeline devel-

opment a capital intensive project (Durust and Joos, 2018). Pipeline infrastructure primarily exists on land, but marine pipelines including deep sea pipeline infrastructure for carbon dioxide transportation is technically feasible. Despite this technical possibility, cost is a barrier to pipelines for transportation across sea. It is less financially viable for storage in marine aquifers due to high costs of offshore installations (Al Baroudi et al., 2021).

Shipping is the second most effective solution for carbon transport given existing technology. Shipping can be for transportation port-to-port or for shipping directly to an offshore injection site (Durust and Joos, 2018). The infrastructure needed to ship CO₂ to an injection site includes temporary storage facilities, loading docks, ships, gasification and liquification technology (Durust and Joos, 2018). Relative to pipeline transportation, shipping costs are lower at further distances as costs lower with the increase in the size of the ship, flexibility in destination and the fixed cost of liquification. In a UK report, shipping costs are estimated to be between £7-£12/tCO₂ (in 2018 £) according to estimates Durust and Joos (2018)

3 Shipping infrastructure elements

Figure 3-1: Components of the CO₂ shipping chain

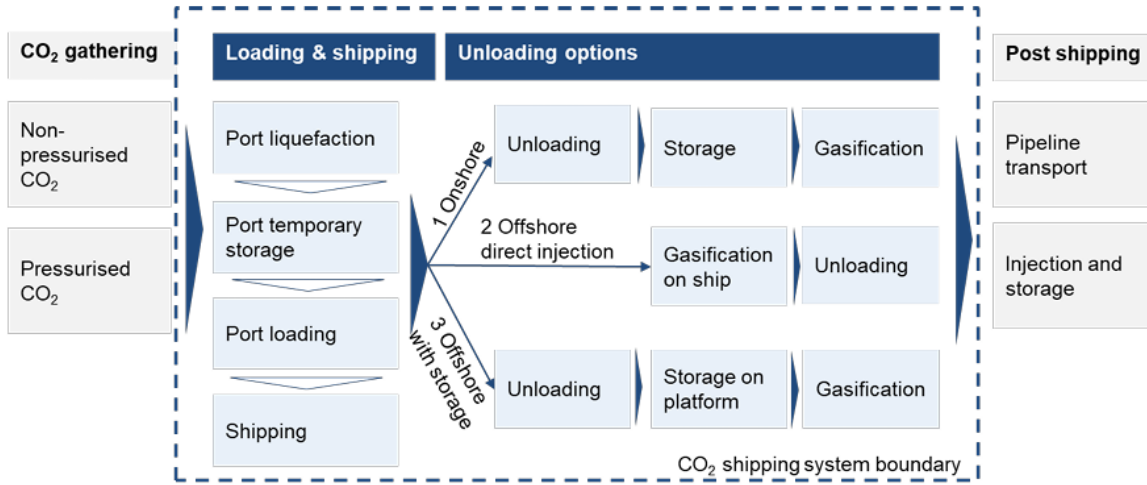


Figure 2: Components of the CO₂ shipping value chain (Durust and Joos, 2018)

- Barges for river transportation also may be possible in the case of transportation needs between smaller installations (Workgroup, 2022). Similarly in Europe, tanks and barges are likely to be useful for transportation of CO₂ from inland sites to seaports (Workgroup, 2022).

- Transportation by rail train is feasible, but there are challenges to its wide-scale deployment. Railway transportation of CO₂ is expensive and has a small carrying capacity (Svensson et al., 2004).
- Trucking is used in CO₂ transportation where CO₂ is moved from a project site to a storage location. Using trucks to transport CO₂ is primarily for small scale projects with nearby storage facilities (Institute, 2018).
- New carbon transportation infrastructure is commonly built in tandem with storage facilities. This is because carbon that is captured for reuse is often reused on site for cost efficiency. CO₂ transportation will be more costly the more distance exists between the site of capture and use or storage (Doctor et al., 2018). Given the few number of storage sites for carbon and a decentralisation of grid infrastructure in many countries, the transportation of carbon risks remaining uneconomical without deep incentives.
- Carbon transportation’s cost and scale is closely linked to the capacity and geography of storage sites. At a high level, pipelines are effective for consistent and high volume of CO₂, while transport via shipping is nonlinearly dependent on distance, and is more ideal for small volumes and long distances (Al Baroudi et al., 2021). Shipping does provide some economies of scale for larger quantities of CO₂, but wanes over time as the cost of liquification of the CO₂ is the highest driver of shipping costs. Pipeline conversion from natural gas to CO₂ will be important for greatly reducing the capital cost of this transportation method. Given how applicable this process is, it is likely existing pipelines connecting industrial and energy generation sites to seaports will be essential for the transportation of CO₂ over long distances to offshore storage sites. Shipping will absolute be important for movement to offshore injection sites as well as port to port transportation for shorter distances. Other forms of transport, including trucking, barges and train are currently less cost effective and not yet scalable, but are likely to play a role in short distances hauls or small scale transportation needs.

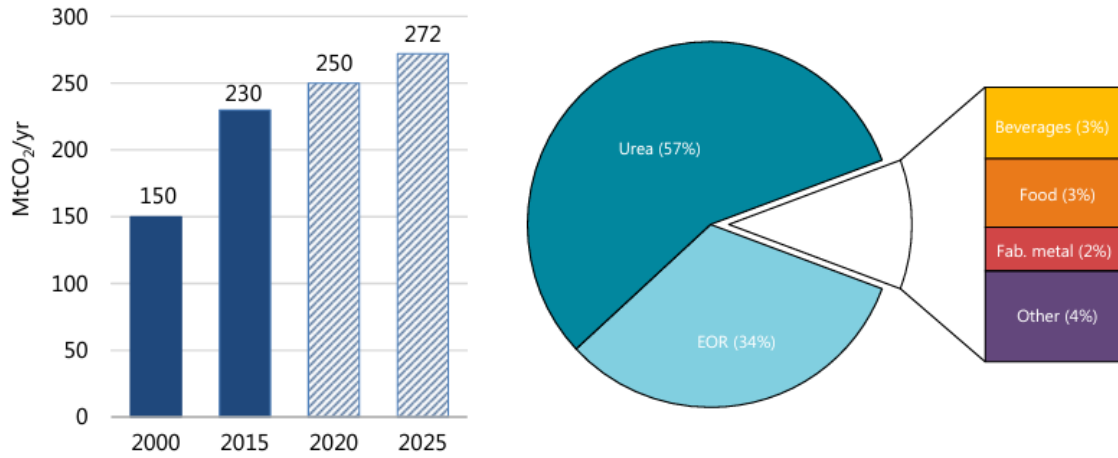
2.3 Carbon utilisation

Carbon utilization is the reintegration or use of captured carbon into industrial processes and technologies as a direct input, feedstock, or solvent. (Commission, 2021). Carbon use depends on the capture of carbon from an industrial or energy generation process and can refer to direct use or indirect use through transformation (IEA, 2023a). It is important to

differentiate the climate benefits of carbon capture and storage versus carbon capture and use. When carbon is used, the net reduction in emissions depends on the source of the CO₂, the energy intensity of converting the CO₂ into another product, and how long that re-used CO₂ remains in the product. In the long run, CO₂ use only provides emissions benefits when sequestered from biomass or the air (IEA, 2023a).

There are different direct uses of carbon once captured, including enhanced oil recovery (EOR), use in fertiliser production - particularly urea production - and food and beverage refrigeration (Serin, 2023). These processes do not require chemical alteration of the CO₂, but may require transformation (i.e. liquification, gasification) and transportation. Classification of direct-use of CO₂ include (1) yield boosting application in fertiliser development, algae growth and greenhouses, (2) solvent creation in EOR, decaffeination of coffee, and dry cleaning, (3) heat transfer fluid in refrigeration and supercritical power systems and (4) other applications in food, beverages, welding and some medical uses (IEA, 2019b).

The US currently leads on number of carbon utilisation projects, with over 17 Mt of CO₂ used annually for EOR, the main purpose of these projects (Bobeck et al., 2019). EOR, done through gas injection of CO₂ to expand an oil reservoir has been cost effective for a number of years, is profitable in a high oil price scenario, and accounts for roughly 60% of US EOR projects (Budzik and Van Wagener, 2014; of Fossil Energy and Management, 2023). In a more detailed analysis, EOR is profitable where there are no geological rifts or cracks in an existing well, historical extraction is known and the cost of a barrel of oil is known (vant Veld and Phillips, 2010). EOR is also a specifically unique use of carbon, given the technology and market have been in existence for four decades. EOR is a particularly unique application of carbon use because of its application predates the demand to lower the carbon intensity of oil production. Contracts for purchasing CO₂ for EOR are normally negotiated by factoring in transportation cost and pegging the per unit cost of CO₂ to the cost of oil (Shen Gao and Zhao, 2022).



Note: Projections for future global CO₂ demand are based on an average year-on-year growth rate of 1.7%.

Sources: Analysis based on ETC (2018), *Carbon Capture in a Zero-Carbon Economy*; IHS Markit (2018), *Chemical Economics Handbook – Carbon Dioxide*; US EPA (2018), *Inventory of U.S. Greenhouse Gas Emissions and Sinks 1990-2016*.

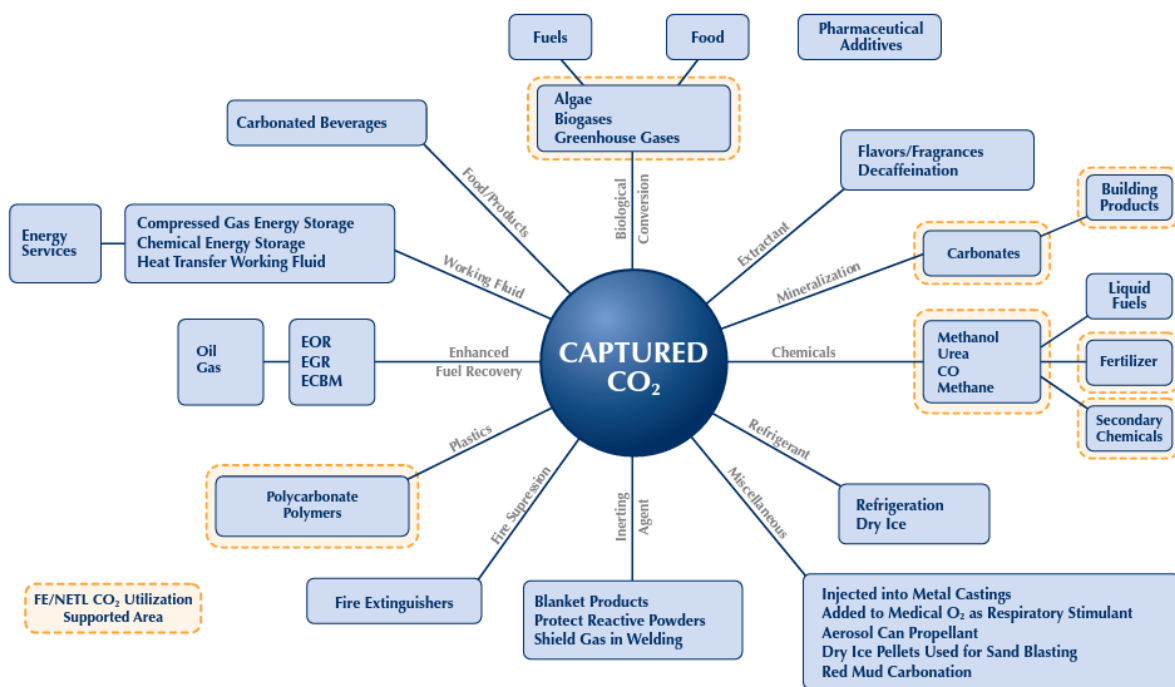
Figure 3: Current and projected demand for CO₂ globally

Different indirect uses of carbon include converting the CO₂ into building materials or plastics (Serin, 2023). These indirect uses, both chemical and biological, are still in development and are very early stages in terms of development. Through chemical and biological conversion, captured CO₂ can be used for (1) creation of fuels including methane, methanol, gasoline, diesel and aviation fuel, (2) chemical production including intermediates and polymers and (3) building materials including cement, concrete and aggregates (IEA, 2019b). Currently, CO₂-based fuels and chemicals have higher production costs than the same products made with fossil fuels, primarily due to the cost of hydrogen production (IEA, 2019b). Utilisation of CO₂ requiring chemical transformation is still, primarily, in the early stages of development with both commercial and regulatory challenges blocking scalability and implementation (IEA, 2019b).

Given the demand for low-carbon building materials, the permanence of carbon storage in the materials and the potential market size, CO₂ use in aggregates, cement and concrete have the most prospect for carbon capture and use projects (Bobeck et al., 2019). CO₂ use in building materials, particularly in cement is currently both cost effective and provides additional climate benefits given the CO₂ remains stored in the cement over the lifetime of its use (IEA, 2019b). The transformation of CO₂ into cement is done through the curing process of cement and enhances the durability, strength and permanently removes the CO₂

from the atmosphere (Li et al., 2022).

The creation of fuels, chemicals and plastics through CO₂ conversion is potentially a large end-use given the demand to lower the emissions intensities of these products in hard-to-abate sectors. CO₂ to fuel and chemical development does require hydrogen, which is currently expensive to produce for industrial processes. Given the need to convert the CO₂, either through steam methane reform (SMR) or electrolysis, production of fuels is currently expensive (Bobeck et al., 2019). But, CO₂ use to create methanol to gasoline is feasible and scalable. Therefore, using CO₂ as a feedstock in a number of chemical production processes is dependent on the technical feasibility of some more difficult processes, but primarily is dependent on the cost of energy and availability of hydrogen to drive down the cost (Bobeck et al., 2019).



Source: National Energy Technology Laboratory www.netl.doe.gov

Figure 4: Captured CO₂ use cases across sectors

Overall, carbon utilisation can deliver on emissions reductions in the near term, but is not a long-term solution for decarbonisation. Use in EOR is likely to rise, even with the growing extraction of gas from shale. Utilisation must be done in tandem with storage, and should only be used in the future (post-2030) where carbon is directly sequestered from the air or via biomass. With the Inflation Reduction Act of 2022, it is likely the US will drive

development and deployment of more carbon utilisation projects ([Serin, 2023](#)).

2.4 Carbon storage

Carbon capture and storage is the only method of permanent removal of CO₂ from the atmosphere back into terrestrial or oceanic systems. Storage of CO₂ is intended to last for thousands of years with minimal leakage. Currently, costs range from US \$61/CO₂ to US \$86/tCO₂ ([Bergstrom and Dyna, 2017](#)).

Storage of CO₂ is needed both temporarily while moving along the transportation supply chain, and for the end use of permanent removal from the atmosphere. Temporary storage of CO₂ is required for transferring CO₂ from either two different states for further transportation or from land to sea. CO₂ storage terminals are necessary for transportation from capture site to storage site and deploy similar LNG storage technologies, as seen at ports and shipping terminals ([Al Baroudi et al., 2021](#)). Temporary storage is not the usual consideration when discussing CO₂ storage, and instead consideration of permanent storage is more widely discussed. Injection of emitted CO₂ back into Earth’s ecosystems is considered an essential component of decreasing the overall stock of CO₂ in the atmosphere. For climate policy, the storage of already emitted CO₂ into geological basins is a key technical solution that can drive a substantial decrease in atmospheric pollutants, without requiring lowering output or increasing efficiency.

In reviewing the components of CCUS technologies, CO₂ storage primarily refers to the process of geological or marine storage as the end use of sequestered carbon. CO₂ storage is the necessary end use component for permanent removal of CO₂ from the atmosphere. Geological carbon storage includes storing extracted CO₂ in oil and gas reservoirs, unminable coal seams, and deep saline reservoirs ([Commission, 2021](#)).

Geological carbon storage is the process of keeping carbon in rock formation or dissolving carbon dioxide into water. Geological storage of this type is required for permanent removal of CO₂ from the atmosphere - i.e. a net reduction in emissions. This can be done mechanically and chemically through a process that involves rock formations, water and minerals. CO₂ storage processes rely on similar physical properties of earth’s rock and soil layers to keep CO₂ securely underground. These include the following four technological processes for geological storage:

- Structural trapping: CO₂ is injected into a rock area and the cap-rock prevents the CO₂ from escaping into the atmosphere. Because CO₂ is bouyant, it is injected deep

into the rock layers but migrates upward until it reaches a layer of impermeable seal rock (of [Fossil Energy and Management, 2023](#)). This mechanism traps the greatest amount of CO₂.

- Residual CO₂ trapping: CO₂ is trapped with capillary forces, where CO₂ enters the porous space in the rock ([CCP, 2021](#)). This will occur about ten years after the CO₂ is injected.
- Solubility trapping (also called dissolution): CO₂ is dissolved into water found within geological formations and sinks. CO₂ dissolves because it is heavy, and the water eventually becomes heavier than normal due to the CO₂, keeping the gas trapped. Lower salinity and water temperatures improve solubility ([Leslie et al., 2021](#)).
- Mineral trapping: This form of CO₂ storage occurs when CO₂ is injected into rock and it binds chemically with the surrounding rock ([CCP, 2021](#)). Mineral trapping cannot be controlled as easily as the other forms of geological storage, and subsequently require mixing the CO₂ with an acidic liquid to accelerate the binding of the CO₂ to the rock minerals ([Hills et al., 2020](#)).

3 Deployment

Based on the disaggregation of technologies provided by the IEA, we look at various configurations of carbon capture, use and storage technologies. These processes are the components, or processes in different technological applications, designed with the aim to remove or reuse atmospheric carbon dioxide emissions. This disaggregation of technologies is demonstrative of the existing ways that the processes are applied, either in commercial use currently or as potential uses. The use of the IEA’s disaggregation of the types of technologies is our most advanced understanding of CCS. The IEA has an explicit and advanced remit for CCS technologies and we will discuss them below.

For each configuration of CCUS technologies, we evaluate both technological efficacy and economic efficiency. Since these technologies involve different sub-processes (as detailed in [Section 2](#)), their deployment and construction result in variations in construction, maintenance, and operational costs. We provide summary statistics for different projects within each category of deployment and compare the maturity of each technology against local carbon prices, where possible.

Technological efficacy, here, refers to the technology’s actual ability to succeed in delivering on the capture, use or storage of carbon dioxide from the atmosphere. Primarily, we focus on the state of the technology in its ability, and focus secondarily on whether the technology will be effective, given economic and policy conditions. We assess efficacy as:

- The technology’s stage of innovation,
- Its scalability or demonstrated ability to scale, and
- The variability in costs, including the factors that influence these variations.

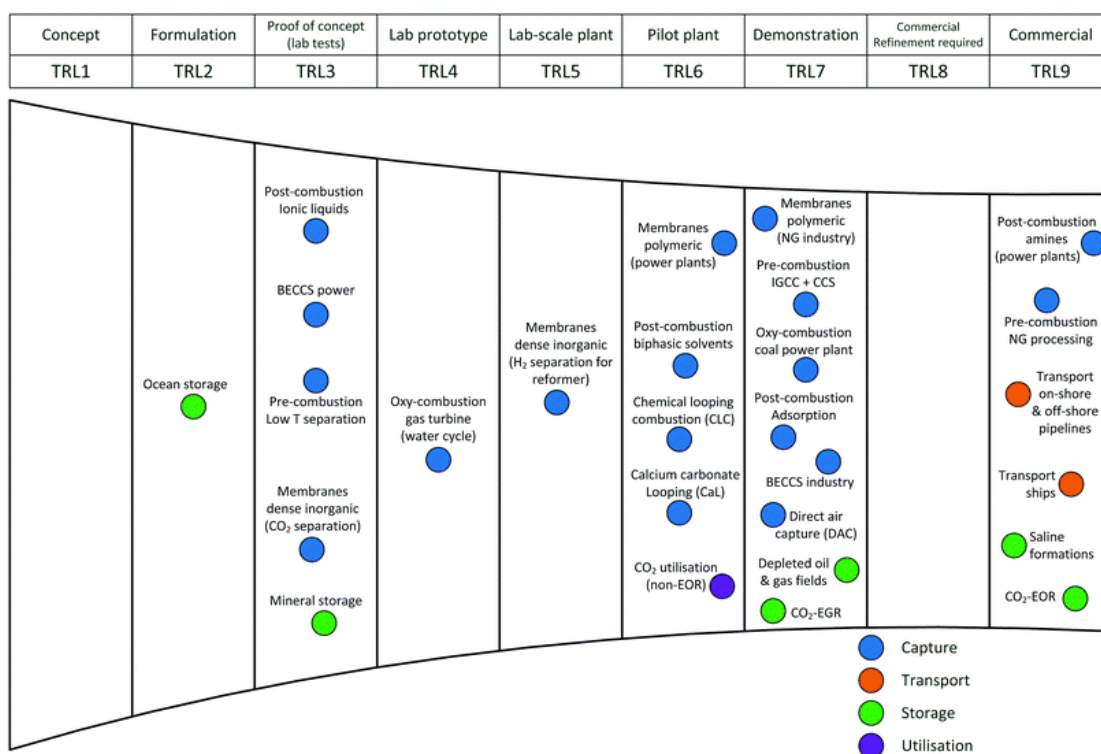


Fig. 1 Current development progress of carbon capture, storage and utilisation technologies in terms of technology readiness level (TRL). BECCS = bioenergy with CCS, IGCC = integrated gasification combined cycle, EGR = enhanced gas recovery, EOR = enhanced oil recovery, NG = natural gas. Note: CO₂ utilisation (non-EOR) reflects a wide range of technologies, most of which have been demonstrated conceptually at the lab scale. The list of technologies is not intended to be exhaustive.

Figure 5: Caption

The economic efficiency of projects will include an estimate of the cost per tonne of carbon for processing carbon dioxide within a broad categorisation of technology. For example,

DACCS is currently the most expensive of the carbon capture, use and storage technologies given its only purpose is to remove carbon dioxide from the atmosphere and there are no price offsets from re-use. Additionally, the carbon is disparate in the atmosphere and therefore requires more power to remove one unit of carbon dioxide relative to directly capturing carbon at a point source of the production of CO₂.

3.1 CCU: direct use

Carbon capture and utilisation is the most developed technology for sequestration of carbon dioxide in the atmosphere. Carbon capture and utilisation (CCU) is often referred to in the context of enhanced oil recovery (EOR). This is a direct use of the captured carbon. A secondary use for carbon sequestered is for re-use in industrial processes. For example, carbon captured can be re-used in the production of a number of chemicals, fertilisers and polymers. Around 230 Mt of carbon dioxide are used annually. Firstly, it is used in the fertiliser industry for urea manufacturing (130 Mt) and for secondly for EOI (about 80 Mt) (IEA, 2023b). Given EOI is a direct application of carbon utilisation, about 75% of installed transmission for carbon dioxide transportation is for EOI (IEA, 2023b).

In terms of net carbon benefits, carbon capture and use has a lower emissions reduction, given the carbon dioxide are reused in industrial processes. Less than 8 Mt of captured CO₂ are used annually for steel, synthetic fuel, fertiliser and chemical production. The market for goods made on recaptured carbon dioxide is still small, primarily constrained by the high cost of carbon capture in the industrial sector. For example, the costs of CO₂ based fuels and chemicals are still much higher than fossil fuel derived substitutes (IEA, 2019b). Given the relatively small scale of application to heavy industry, compared with other technologies, it is not widely seen as a driver of net emissions reductions among the varying technologies in the CCUS space.

3.2 CCU: indirect use

All processes which chemically alter recaptured carbon dioxide are indirect use applications. For example, conversion of CO₂ into fuels, including methane, methanol and aviation fuels is possible IEA (2023a). Other applications in the building sector include conversion of CO₂ into aggregates, cement and concrete, as well as in the chemical sector, where captured CO₂ is converted to chemical intermediates.

Indirect use of captured CO₂ is most applicable in the industrial sector. In terms of tech-

nological readiness, existing chemical conversion processes and infrastructure in the heavy industry sector has made it possible to implement. But, costs remain high.

3.3 CCUS: storage in industry

[Naucner and Ruijs \(2008\)](#) suggest applications of CCS to coal power plants will be less expensive to industrial applications, although rival analysis indicates it is more costly. The main driver of the cost is the actual carbon capture technology, as opposed to the transportation and storage costs. In a rising carbon price market and rapidly shifting energy mix in electricity sector, it is likely the economics of CCS for coal are becoming more difficult. For example, in 2008, coal comprised 30% of the EU's energy mix, while today, it constitutes only 15% [Union \(2023\)](#). Cost estimates from reports in 2011 indicate XX per average ton of the [European Technology Platform for Zero Emission Fossil Fuel Power Plants \(2011\)](#).

Sectors of deployment matched to the above technologies (for instance, the power sectors uses CCS/CCUS, and the chemical sector uses CCS). Give strong description of how each of these works.

Power sector The power sector deploys the largest number of CCUS projects. Over 60% of operating CCUS projects are for natural gas processing ([IEA, 2019a](#)). Removal of CO₂ from the gas stream is a necessary component of natural gas processing, making it the least cost form of CCUS, at 15 to 20 USD per tonne (?). In 2021, 28.5 Mt of CO₂ were sequestered from currently active natural gas processing projects.

- Industrial sector

3.4 CCS: geological storage

- Geological storage - DAC sector

3.5 Bioenergy with carbon capture and storage

3.6 Direct air capture

Without DAC, the need to decarbonise the electricity sector by 2030 is urgent. 70% of energy emissions must be cut by 2030, meaning intermittent renewables would be necessary to drive all energy system emissions reductions in absence of DAC ([Realmonte et al., 2019](#)).

Expert survey results suggest DAC or CDR technological feasibility is highly uncertain, despite model reliance on the technology ([Grant et al., 2021](#)).

Two types of DAC that are potentially feasible. There are solid (S-DAC) and liquid (L-DAC) technologies. S-DAC requires more energy, and operates by creating a vacuum to absorb carbon while L-DAC is heat intensive and requires a water-based solution and high temperatures to capture carbon ([IEA, 2023c](#))

The US has announced a suite of energy prizes for different technologies at different stages under the American Made funding. Recently, early stage DAC prize was announced for \$35 million ([HeroX, 2023](#); [of Energy, 2023](#)).

4 Policy environment

An early report by McKinsey & Company in tandem with the Global CCS Institute published some of the first figures on the economics of the technology at large scale deployment in the industry. Cost modelling from 2008 data predicted a drop in CCS prices to 30 to 45 euro per tonne by 2030, modelled in line with the EU Emissions Trading Scheme (ETS) prices [Naucner and Ruijs \(2008\)](#). ETS credits have been trading at around 80 euro per tonne since 2021 after a steep rise from 25 euro in 2020 with the move into Phase VI of the ETS scheme and a restriction of credit supplies. In today's rapidly developing market, what lessons are still relevant? What we do know is the price of carbon in Europe remains high, and will continue as more ambitious policies, in addition to the carbon trading scheme, remain in place. Whether this will be enough to drive an acceleration in the adoption of CCS technology remains uncertain.

Going back to previous estimates of CCS costs and the price trajectory over time can indicate the magnitude of barriers to technological deployment. As suggested, [Naucner and Ruijs \(2008\)](#) determined global abatement could reach 3.6 Gt of emissions by 2030 cost effectively, while the [IEA \(2023a\)](#) suggests, even with the existing pipeline of projects, global abatement through CCS will fall short of the 1.4 Gt target they have developed under their Net Zero Emissions model. Projects are currently not being delivered either at a low enough cost or a wide enough scale to meet the necessary levels of deployment for mitigation of CO₂ emissions as urged by scientists.

Despite the lack of progress on CCS deployment on average, there is sectoral level variation across industries. The main use of CCS will be seen in the energy sector, the industry sector (cement, steel and other heavy manufacturing), and other manufacturing including chemical production.

- COP28 potential for CCUS/CCU growth. Also, general past COPs have had a lean

towards developing these technologies more effectively – IPCC / IEA / CCC reports all rely on the deployment of CCS to meet minimum carbon removal goals, cannot just be forests

- According to the IEA Sustainable Development Scenario, [begin quote] in which global CO₂ emissions from the energy sector fall to zero on a net basis by 2070, CCUS accounts for nearly 15% of the cumulative reduction in emissions compared with the *Stated Policies Scenario*. [end quote]
- Regional: US (IRA and new EPA ruling for power plants), EU (Horizons), Middle East (looking at current regional funding programmes and also from the IEA data on projects).
- US ambitions and prizes
- IEA policy database

4.1 Carbon markets

- Carbon pricing and DAC - map the current projects and cost per tonne of CO₂e relative to the cost per CO₂e in the VCM (in absence of a compliance market). We can map the feasibility of a market for DAC and what prices are needed for DAC to be deployable

- Expansion of the VCM and carbon pricing to see how this could also feature on developing CCS projects. but, the overall idea is that VCM offers a market price for carbon, but it is significantly lower than regulatory prices. Also, VCM is a different tool, it does not offer the early stage R&D support that may be necessary to create a scalable technology that is more on the upper side of the innovation curve. Looking at the case for China this could be important.

5 Economics

5.1 Advanced market commitments

An advanced market commitment (AMC) is an investment vehicle in order to fund research with a high public benefit and low private incentives.

We look for a current example. Frontier, a private US-based firm, has created a fund comprised of monetary commitments by other private firms in order to buy units of carbon

offsets generated exclusively via CCS technologies. To date, the fund has secured over one billion USD in funding from large corporations such as Stripe, McKinsey and Microsoft. This is one prime example of a large private agent creating a mechanism to induce or incentivize innovation in a technology that has potentially high welfare gains despite being unprofitable at this time.

The fund created by Stripe is advertised as an AMC, where committed dollars are then dispensed to qualifying firms dotted along the innovation ladder, at Frontier’s discretion. These CCS technology producing firms have agreed to provide a predetermined amount of offset credits at a given time in the future. We leave a more full discussion of the implementation of the mechanism by Frontier until later in the text, focusing now on the conceptual design of an AMC.

AMCs were originally studied for vaccine delivery [Kremer et al. \(2020, 2022\)](#). Earliest economics research into AMCs by [\(Glennerster et al., 2006\)](#) depict the economic intuition and a potential application in healthcare, for vaccines. A brief review of the economic motivation and mechanism for AMC was discussed in [\(Kremer et al., 2020\)](#). Their first in kind assessment was a model depicting altruistic donations by national governments. They assumed governments would contribute through an AMC to accelerate the development of a vaccine. The authors find the hold up problem in investments can be deterred through the AMC mechanism. AMCs can help recoup funding of large investments where profits may not be sufficient (low-income markets), and technically distant (long-shot) technologies. Despite the strength of the model, no counterfactual was available to test this hypothesis, but authors assume the benefit is large relative to the costs of the AMC mechanisms. Overall research indicates the hold-up problem is likely to be resolved [Kremer et al. \(2022\)](#).

5.2 Research and development

Research and development subsidies (R&D) subsidy effectiveness is not clear, there is limited evidence in the empirical literature. [Howell \(2017\)](#) investigates the use of early stage research and development funding in the United States, on the impacts of innovation. There are two channels that R&D can provide for support for innovation:

1. compensating for the underinvestment in innovative technologies due to disparate social benefits;
2. reducing financial frictions.

Most successful or high impact of the R&D subsidy is for early-stage prototyping. The R&D grant is able to provide enough funding to allow the firm to invest in demonstrative technologies, attracting private funding in the future. There is proof that early-stage R&D funding can support clean innovation by exploiting discontinuity at the thresholds.

5.3 Directed technical change

Currently, there exists a gap in literature on the particular economics of AMCs for long shot mitigation technologies. Despite this gap, there is some early work motivated by the technical change framework for how CCS technologies are likely to develop in the near future. (Grimaud and Roug  , 2019) use a computational general equilibrium model to evaluate future emissions scenarios with the option to adopt CCS in renewable energy sector. Model emissions are endogenous, and CCS is subsequently adopted to mitigate, where the option is less expensive. Model findings suggest there is an underprovision of CCS in the near term, due to optimal emissions path ambiguity. For some countries, it may be optimal to have more growth, and with that more extraction and emissions in the near term. This model does not account for directed technical change, so results are limited in their application.

Next, we review the directed technical change (DTC) literature as a method for lowering the cost of mitigation technologies. (Acemoglu et al., 2012) review directed technical change and environmental outcomes. Using a model of clean and polluting technologies, the authors evaluate, under different policy scenarios, the adoption of clean technologies. Their findings suggest a delay in the transition is costly, as intervention will occur in the dirty sectors until the price of pollution is internalised (i.e. carbon tax). In spite of this, polluting sectors are tied to exhaustible resources, which means even in a laissez-faire scenario, clean innovation will occur. Other results are important for our understanding of innovation in mitigating technologies, including the existence of path dependence in the development of clean versus dirty technologies. Additionally, innovation is endogenous, warranting calls for earlier action. And finally, in addition to Pigovian carbon taxation, the optimal policy includes research subsidies specifically devoted to clean innovation. These findings suggest, while not explicit, large scale and early subsidisation of clean technologies may allow for quicker adjustments to a low-emissions pathway.

A recent meta-analysis by H  mous and Olsen (2021) has reviewed the literature on DTC in application to the environment and labour. With respect to environment, the summary of the literature suggests climate policy should focus on innovation in the near term, given the gap in emissions reductions and evidence to suggest the damages from emissions along

our current trajectory. But, [Hémous and Olsen \(2021\)](#) find the distribution of social returns to any clean innovation will then impact further industrial innovation policies. The first set of innovation policies can have knock-on effects for future policy development.

6 Case study

Text ...

7 Conclusions

Costs are currently high. CCUS applications do not all have the same cost. Looking specifically at carbon capture, the cost can vary greatly by CO₂ source, from a range of USD 15-25/t CO₂ for industrial processes producing pure or highly concentrated CO₂ streams (such as ethanol production or natural gas processing) to USD 40-120/t CO₂ for processes with dilute gas streams, such as cement production and power generation. Capturing CO₂ directly from the air is currently the most expensive approach, but could nonetheless play a unique role in carbon removal. Some CO₂ capture technologies are commercially available now, while others are still in development, and this further contributes to the large range in costs. There is considerable potential to reduce costs along the CCUS value chain, particularly as many applications are still in the early stages of commercialisation. Experience indicates that CCUS should become cheaper as the market grows, the technology develops, finance costs fall, economies of scale are reached, and experience of building and operating CCUS facilities accumulates. This pattern has already been seen for renewable energy technologies over recent decades.

Cost reductions have already been achieved at large-scale CCUS projects. For example, the cost of CO₂ capture in the power sector has come down by 35% through its evolution from the first to the second large-scale CCUS facility, and this trend is set to continue as the market expands. There is considerable potential to reduce costs along the CCUS value chain, particularly as many applications are still in the early stages of commercialisation. Experience indicates that CCUS should become cheaper as the market grows, the technology develops, finance costs fall, economies of scale are reached, and experience of building and operating CCUS facilities accumulates. This pattern has already been seen for renewable energy technologies over recent decades. Cost reductions have already been achieved at large-scale CCUS projects.

More public funding can support cost-reducing innovations. The relative lack of progress in deploying CCUS to date means that many technologies and applications are still at an early stage of commercialisation, and therefore at a high point in the cost curve. There is ample potential for cost reductions. The experience of wind and solar highlights what is possible â but, as with renewable energy, realising this potential will require strengthened policy support to drive CCUS innovation and deployment.

Grant funding programmes can play an important role in supporting early CCUS deployment, particularly first-of-a-kind projects and CO₂ transport and storage infrastructure. They can alleviate the high capital costs and commercial and technical risks associated with such projects. However, they can be a heavy burden on public budgets, limiting their use to a small number of individual facilities. A rapid scale-up of CCUS will necessitate a shift towards market-based measures that can complement grant funding and provide a stable and ongoing framework for CCUS facilities to operate over the long term

Increased policy clarity can improve alignment with private incentives In the pursuit of net zero, we cannot afford to dismiss CCUS as "too expensive". It is the only group of technologies that can contribute both to reducing emissions in critical economic sectors and to removing CO₂ to balance emissions that cannot be avoided, a balance that is at the heart of net-zero ambitions. In some sectors, including heavy industry, CCUS is currently the least-cost or only practical option for deep emissions reductions.

CCUS offers an opportunity for policy experimentation:

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Figures and tables

A Appendix

TBA...